

🔥 LiquidAI / **LFM2.5-350M** 📄

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4.06k



Text Generation



Transformers



Safetensors



9 languages

lfm2

liquid

lfm2.5

edge

conversational



Eval Results



arxiv:2511.23404



License: lfm1.0



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Use this model ▾



Model card



Files



xet



Community

7



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LFM2.5-350M

LFM2.5 is a new family of hybrid models designed for **on-device deployment**. It builds on the LFM2 architecture with extended pre-training and reinforcement learning.

- **Best-in-class performance:** A 350M model rivaling much larger models, bringing high-quality AI to your pocket.
- **Fast edge inference:** 313 tok/s decode on AMD CPU, 188 tok/s on Snapdragon Gen4. Runs under 1GB of memory with day-one support for llama.cpp, MLX, and vLLM.
- **Scaled training:** Extended pre-training from 10T to 28T tokens and large-scale multi-stage reinforcement learning.

Downloads last month

74,543



Safetensors ⓘ

Model size 0.4B params

Tensor type BF16

Chat template

Files info

⚡ **Inference Providers** **NEW**

Text Generation

This model isn't deployed by any Inference Provider.



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Ask for provider support

Model tree for LiquidAI/LFM2.5-3...

Base model LiquidAI/LFM2.5-350M-Ba...



Finetuned (13) [this model](#)

Adapters 14 models

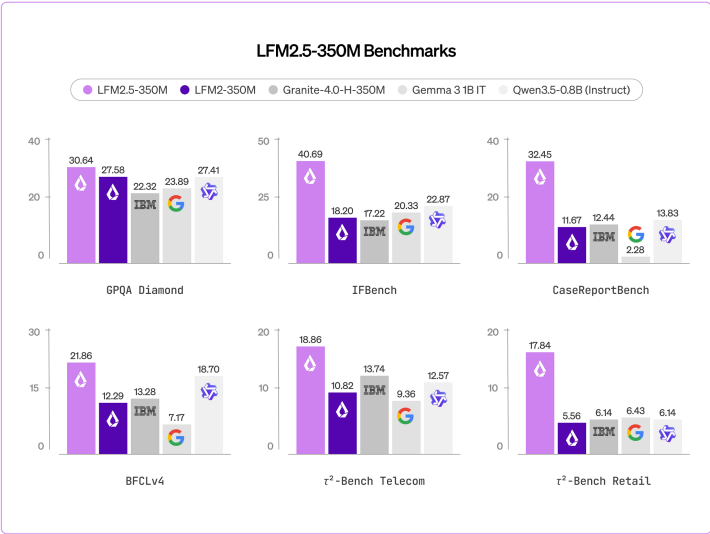
Finetunes 45 models

Quantizations ⚡ 📄 🙌 38 models

Spaces using LiquidAI/LFM2.5-... 7

Find more information about LFM2.5-350M in our [blog post](#).

 **Demo:** <https://huggingface.co/spaces/webml-community/lfm2.5-webgpu-summarizer>



Model Details

Model	Parameters	Description
LFM2.5-350M-Base	350M	Pre-trained base model for fine-tuning
LFM2.5-350M	350M	General-purpose instruction-tuned model

LFM2.5-350M is a general-purpose text-only model with the following features:

- **Number of parameters:** 350M
- **Number of layers:** 16 (10 double-gated LIV convolution blocks + 6 GQA blocks)
- **Training budget:** 28T tokens

-  [webml-community/lfm2.5-webgpu-s...](#)
-  [FlameFOX/SLM-Playground](#)
-  [Carnot-EBM/hallucination-detector](#)
-  [SrhKsn/lfm2-mentalchat-demo](#)
-  [FlameFOX/lfm2](#)
-  [tushman/cap-6640-assignment-4](#)
-  [ushort/uncensored](#)

 **Collection including LiquidAI/LFM...**

 **LFM2.5**  Collection

Collection ... • 14 items • Up... •  168

 **Paper for LiquidAI/LFM2.5-350M**

LFM2 Technical Report

 Paper • 2511.23404 • Publi... •  65

Evaluation results

- [ldavidrein/gpqa · Diamond](#)   30.64
- [TIGER-Lab/MMLU-Pro · Mml...](#)   20.01

- **Context length:** 32,768 tokens
- **Vocabulary size:** 65,536
- **Knowledge cutoff:** Mid-2024
- **Languages:** English, Arabic, Chinese, French, German, Japanese, Korean, Portuguese, Spanish
- **Generation parameters:**
 - temperature: 0.1
 - top_k: 50
 - repetition_penalty: 1.05

Model	Description
<u>LFM2.5-350M</u>	Original model checkpoint in native format. Best for fine-tuning or inference with Transformers and vLLM.
<u>LFM2.5-350M-GGUF</u>	Quantized format for llama.cpp and compatible tools. Optimized for CPU inference and local deployment with reduced memory usage.
<u>LFM2.5-350M-ONNX</u>	ONNX Runtime format for cross-platform deployment. Enables hardware-accelerated inference across diverse environments (cloud, edge, mobile).
<u>LFM2.5-350M-MLX</u>	MLX format for Apple Silicon. Optimized for fast inference on Mac devices using the MLX framework.
<u>LFM2.5-350M-OpenVINO</u>	OpenVINO format for Intel hardware acceleration. Optimized for efficient inference on Intel CPUs, GPUs, and NPUs.

We recommend using it for data extraction, structured outputs, and tool use. It is not recommended for knowledge-intensive tasks and programming.

Chat Template

LFM2.5 uses a ChatML-like format. See the [Chat Template documentation](#) for details. Example:

```
<|startoftext|><|im_start|>system
You are a helpful assistant trained by Liquid
<|im_start|>user
What is C. elegans?<|im_end|>
<|im_start|>assistant
```

You can use `tokenizer.apply_chat_template()` to format your messages automatically.

Tool Use

LFM2.5 supports function calling as follows:

1. **Function definition:** We recommend providing the list of tools as a JSON object in the system prompt. You can also use the `tokenizer.apply_chat_template()` function with tools.
2. **Function call:** By default, LFM2.5 writes Pythonic function calls (a Python list between `<|tool_call_start|>` and `<|tool_call_end|>` special tokens), as the assistant answer. You can override this behavior by asking the model to output JSON function calls in the system prompt.

3. **Function execution:** The function call is executed, and the result is returned as a "tool" role.
4. **Final answer:** LFM2 interprets the outcome of the function call to address the original user prompt in plain text.

See the [Tool Use documentation](#) for the full guide.
Example:

```
<|startoftext|><|im_start|>system
List of tools: [{"name": "get_candidate_status"}]
<|im_start|>user
What is the current status of candidate ID 12345?
<|im_start|>assistant
<|tool_call_start|>[get_candidate_status(candidate_id="12345")]
<|im_start|>tool
[{"candidate_id": "12345", "status": "Interview scheduled"}]
<|im_start|>assistant
The candidate with ID 12345 is currently in the interview stage.
```

 Inference

LFM2.5 is supported by many inference frameworks.
See the [Inference documentation](#) for the full list.

Name	Description	Docs	Notebook
Transformers	Simple inference with direct access to model internals.	Link	 Open in Colab
vLLM	High-throughput production deployments with GPU.	Link	 Open in Colab

Name	Description	Docs	Notebook
llama.cpp	Cross-platform inference with CPU offloading.	Link	
MLX	Apple's machine learning framework optimized for Apple Silicon.	Link	—
LM Studio	Desktop application for running LLMs locally.	Link	—
OpenVINO	Intel's toolkit for optimized inference on CPUs, GPUs, and NPUs.	Link	—

Here's a quick start example with Transformers:

```
from transformers import AutoModelForCausalLM

model_id = "LiquidAI/LFM2.5-350M"
model = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map="auto",
    dtype="bfloat16",
    # attn_implementation="flash_attention_2"
)

tokenizer = AutoTokenizer.from_pretrained(model_id)
streamer = TextStreamer(tokenizer, skip_prompt=True)

prompt = "What is C. elegans?"

input_ids = tokenizer.apply_chat_template(
    [{"role": "user", "content": prompt}],
    add_generation_prompt=True,
```

```
        return_tensors="pt",
        tokenize=True,
    )["input_ids"].to(model.device)

    output = model.generate(
        input_ids,
        do_sample=True,
        temperature=0.1,
        top_k=50,
        repetition_penalty=1.05,
        max_new_tokens=512,
        streamer=streamer,
    )
```

Fine-Tuning

We recommend fine-tuning LFM2.5 for your specific use case to achieve the best results.

Name	Description	Docs	Notebook
CPT (Unsloth)	Continued Pre-Training using Unsloth for text completion.	Link	 Open in Colab
CPT (Unsloth)	Continued Pre-Training using Unsloth for translation.	Link	 Open in Colab
SFT (Unsloth)	Supervised Fine-Tuning with LoRA using Unsloth.	Link	 Open in Colab
SFT (TRL)	Supervised Fine-Tuning with LoRA using TRL.	Link	 Open in Colab
DPO (TRL)	Direct Preference Optimization with LoRA using TRL.	Link	 Open in Colab

Name	Description	Docs	Notebook
GRPO (Unsloth)	GRPO with LoRA using Unsloth.	Link	 Open in Colab
GRPO (TRL)	GRPO with LoRA using TRL.	Link	 Open in Colab

Performance

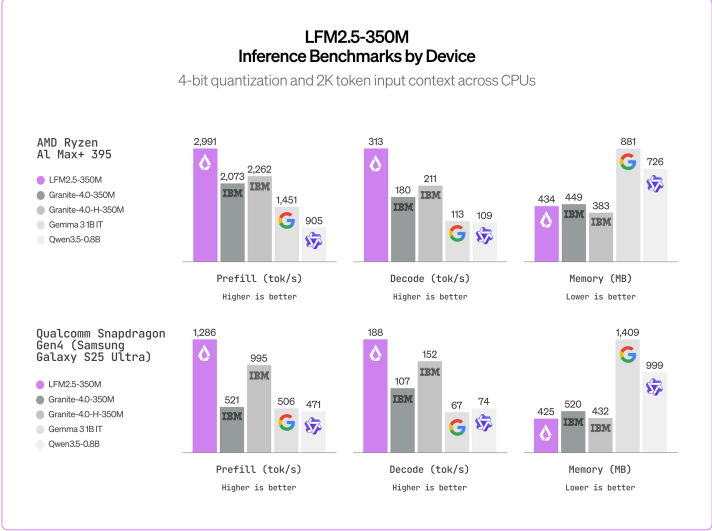
Benchmarks

Model	GPQA Diamond	MMLU-Pro	IFEval	IFBench	Multibench IF
LFM2.5-350M	30.64	20.01	76.96	40.69	44.90
LFM2-350M	27.58	19.29	64.96	18.20	32.90
Granite 4.0-H-350M	22.32	13.14	61.27	17.22	28.70
Granite 4.0-350M	25.91	12.84	53.48	15.98	24.20
Qwen3.5-0.8B (Instruct)	27.41	37.42	59.94	22.87	41.60
Qwen3.5-0.8B (Thinking)	19.29	-*	32.93	22.00	26.40
Gemma 3 1B IT	23.89	14.04	63.49	20.33	44.20

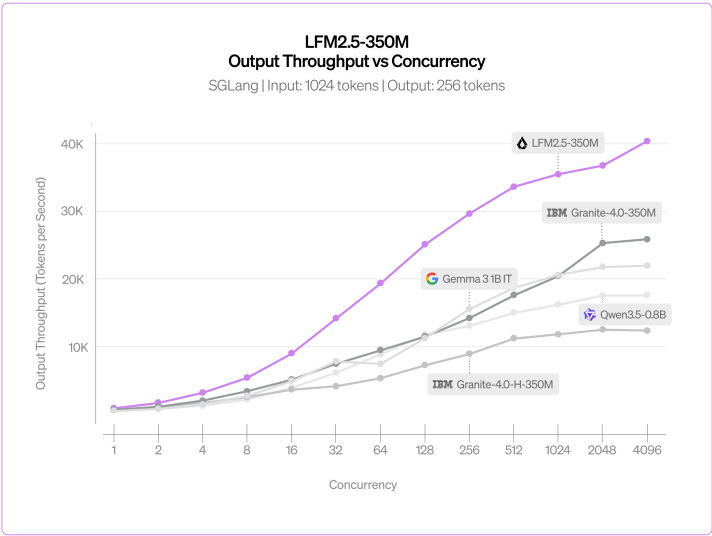
Model	CaseReport Bench	BFCLv3	BFCLv4	τ Be Tele
LFM2.5-350M	32.45	44.11	21.86	18.8
LFM2-350M	11.67	22.95	12.29	10.8
Granite 4.0-H-350M	12.44	43.07	13.28	13.7
Granite 4.0-350M	0.84	39.58	13.73	2.92
Qwen3.5-0.8B (Instruct)	13.83	35.08	18.70	12.5
Qwen3.5-0.8B (Thinking)	0.39	39.64	25.39	14.3
Gemma 3 1B IT	2.28	16.61	7.17	9.36

**Evaluation could not be completed due to doom looping.*

CPU Inference



GPU Inference



Contact

- Got questions or want to connect? [Join our Discord community](#).
- If you are interested in custom solutions with edge deployment, please contact [our sales team](#).

Citation

@article{liquidAI2026350M,

System theme liquid AI, Privacy

About

Careers



Models

Datasets

Spaces

Pricing

Docs

```
title = {LFM2.5-350M: No Size Left Behind}  
journal = {Liquid AI Blog},  
year = {2026},  
note = {www.liquid.ai/blog/lfm2-5-350m-no-:  
}
```

```
@article{liquidai2025lfm2,  
title={LFM2 Technical Report},  
author={Liquid AI},  
journal={arXiv preprint arXiv:2511.23404},  
year={2025}  
}
```